



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

UNIVERSITY EXAMINATION 2025/2026

THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF  
BACHELOR OF SCIENCE MECHATRONIC ENGINEERING

EMT 3104: MECHATRONIC SYSTEM PROGRAMMING I

DATE: 11<sup>TH</sup> AUGUST 2025

TIME: 08.30-10.30AM

**Instructions**

1. This paper contains FIVE questions
2. Answer Question **ONE** (Compulsory Question), **ONE** question in **Section A** (**EITHER** Question TWO or Question THREE) and **ONE** question in **Section B** (**EITHER** Question FOUR or Question FIVE)
3. Remember to indicate all steps and the relevant units in any calculations

**Question ONE (Compulsory) [30 marks]**

- a) A function is a means of creating reusable segments of code making programs more readable. Write a function with the following declaration format:  
`int smallestNum(int num1, int num2, int num3)` that takes in 3 integer numbers and returns the smallest of the 3 using if else if statements. **[4 Marks]**
- b) A bottling company is upgrading its conveyer- system input system. The new sensors have an input voltage range of 0v–10v. If the system Analogue to Digital Converter (ADC) is 12 bits, how many quantized levels are available and what is the resolution of the ADC. **[4 Marks]**
- c) The ATmega32 has a feature known as a **Timer**. Discuss their functionality and give examples for their application. **[3 Marks]**
- d) **Macros** and the **Ternary operator** are key components of writing modular C programs. Below is a sample program that implements both concepts.
  - 1) Highlight the section of the code that implement macros and the ternary operator, explaining their functionality within the code **[3 Marks]**
  - 2) Show the output of the compiler when the code below is run, explaining how you arrived at your result. **[3 Marks]**

```

1. #include <stdio.h>
2. #define square(x) (x*x)
3. #define divisible(x) (x % 2 == 0)
4. int main()
5. {
6.     int input = 3;
7.     int result = 15/square(input);
8.     int check = divisible(result) ? 1:0;
9.     if (check){
10.        printf("%d, is okay", result);
11.    }else{
12.        printf("%d, needs checking", result);
13.    }return 0;
14. }

```

- e) Pulse width modulation (PWM) is a technique used to create a varying voltage signal from a microcontroller system. Using a suitable diagram, explain how PWM signal is achieved and how the duty cycle is calculated. **[5 Marks]**
- f) Field programmable logic controllers (FPGA) and programmable logic controllers (PLC) are alternative programmable controllers to typical microcontrollers. State 3 advantages of each and a typical application where they can be used. **[8 marks]**

### SECTION A

#### Question TWO [20 marks]

- a) A macro is a rule or code pattern that specifies how a certain input should be mapped to a certain output. Highlight the 2 main types of C programming macros with suitable examples for each. **[6 Marks]**
- b) Using a well-illustrated diagram, discuss the startup process of a typical microcontroller program. **[8 Marks]**
- c) The Inter-integrated Circuit (I<sup>2</sup>C) Protocol is a widely used communication protocol for embedded systems. Using a suitable diagram shows typical device connections involved in I<sup>2</sup>C and highlights its key distinguishing features as compared to UART and SPI. **[6 marks]**

#### Question THREE [20 Marks]

- a) Calculate the Hexadecimal value that should be stored in TCNT0 register to achieve a timer0 delay of 20  $\mu$ s with an operating frequency of Fosc = 16MHz. Available prescaler values are (1, 8, 64 and 256). **[6 Marks]**
- b) Using a well labelled diagram, define what is meant by Serial Peripheral Interface (SPI) and highlight the communication pins used and their purpose during communication. **[6 Marks]**
- c) When interfacing digital inputs e.g. push buttons, there is need to handle noisy inputs using techniques such as debouncing. Using a suitable diagram explain the main source of digital input noise for such inputs and discuss examples of hardware and software debouncing techniques. **[8 Marks]**

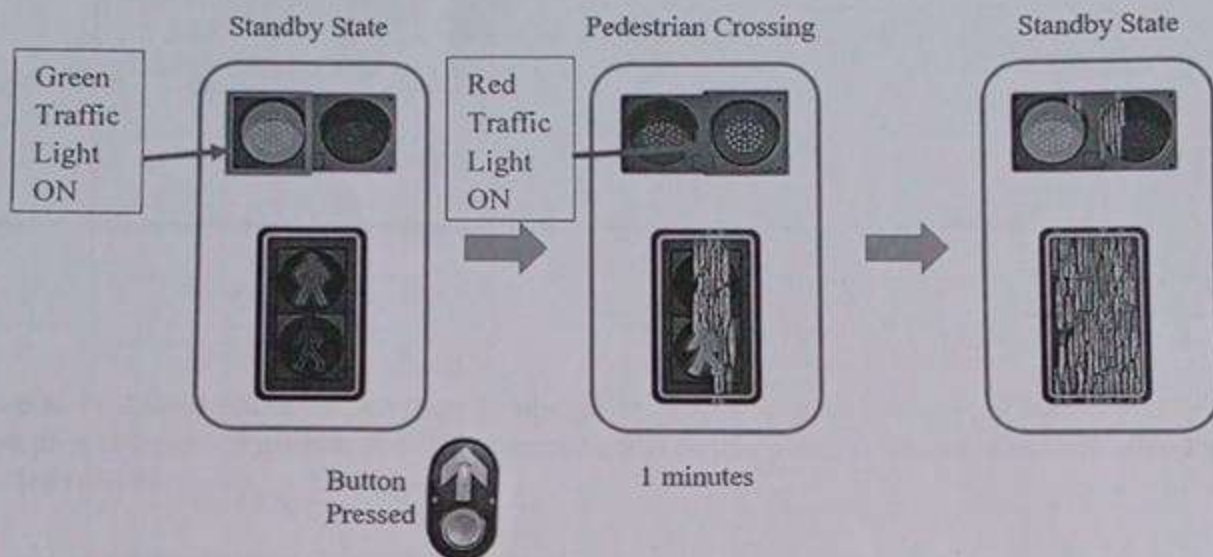


## SECTION B

### Question FOUR [20 Marks]

- a) Interrupts are a very useful concept when dealing with microcontrollers, using a well-labelled diagram, explain the term Interrupt, in relation to microcontroller programming and how they operate. Discuss the 2 distinct types and some common applications where they are utilized. [6 Marks]
- b) You have been contracted as a system designer to create a working mechatronic system to control a traffic /pedestrian crossing intersection as shown in **Fig Q4b**. There are 2 lights (Red and Green) for controlling the traffic and 2 lights (Red and Green) for controlling pedestrians. There is a Pedestrian button which pedestrians can press when the need to cross the street. The system is to be modelled using 4 LEDs (1 Red and 1 Green for traffic lights connected to PB0 and PB1) and (1 Red and 1 Green for pedestrian lights connected to PB2 and PB3). The Pedestrian button will be modelled using a simple push button connected to PC0

*PC0 - Red*



**Figure Q4b: Traffic-pedestrian crossing control lights.**

**System Operation:** During normal operation, the traffic light should be Green and the Pedestrian light should be Red (Standby State). When a pedestrian arrives at the crossing and presses the Pedestrian button, the Green traffic light should turn Red to halt the flow of traffic. The Red Pedestrian light should turn Green to allow pedestrians to cross the road. The Red traffic light and Green pedestrian lights should remain on for 1 minute and then return to their original states i.e. Red traffic light returns to Green and the Green pedestrian light returns to Red until the next time the Pedestrian button is pressed.

- i) Draw a suitable circuit based on the above specifications. [3 Marks]
- ii) Write a suitable flowchart. [3 Marks]

- iii) Write a suitable executable C program, using the given template as a guide, that will run on the ATmega32. [8 Marks]

```
/* Include header files */
#define F_CPU 8000000UL
#include <avr/io.h>
#include <util/delay.h>

/* Function Declaration */
void Pedestrian_cross();
void Standby_State();

int standby = 0;
int main(void)
{
    /* Pin Declaration */
    while(1)
    {
        /* Read status of pin PC0 */
        /* If the button is pressed and the system is not in standby.*/

    }
    return 0;
}

/* Function to turn traffic light RED and Pedestrian light GREEN */
void Pedestrian_cross() {
}

/* Function to turn traffic light GREEN and Pedestrian Light RED */
void Standby_State() {
}
```

### Question Five [20 Marks]

- a) Below is a simple C program used to display the grade of students, write the flowchart of the code below and the expected console output when the code is run assuming it complies with no errors. [5 Marks]

```
#include <stdio.h>

int main () {

    /* local variable definition */
    char grade = 'B';

    switch(grade) {
        case 'A' :
            printf("Excellent!\n");
            break;
        case 'B' :
            printf("Well done\n");
            break;
        case 'C' :
            printf("You passed\n");
            break;
        case 'D' :
            printf("Better try again\n");
            break;
    }
}
```



```

    default :
        printf("Invalid grade\n");
    }

    printf("Your grade is %c\n", grade);
    return 0;
}

```

b) Below is a simple connection circuit for operating a 7-segment display and its associate table of HEX numbers as shown in Fig Q5b. You are instructed to develop a compliable C program to drive the display while displaying 9 to 0 with an interval of 1000 ms between displaying each number.

- Develop a suitable flowchart for this application. [5 Marks]
- Using the template below write the C program for accomplishing the task. [10 Marks]

|   | Binary   | HEX  |
|---|----------|------|
| 0 | 00111111 | 0x3F |
| 1 | 00000110 | 0x06 |
| 2 | 01011011 | 0x5b |
| 3 | 01001111 | 0x4f |
| 4 | 01100110 | 0x66 |
| 5 | 01101101 | 0x6d |
| 6 | 01111101 | 0x7d |
| 7 | 01110000 | 0x70 |
| 8 | 01111111 | 0x7f |
| 9 | 01101111 | 0x6f |

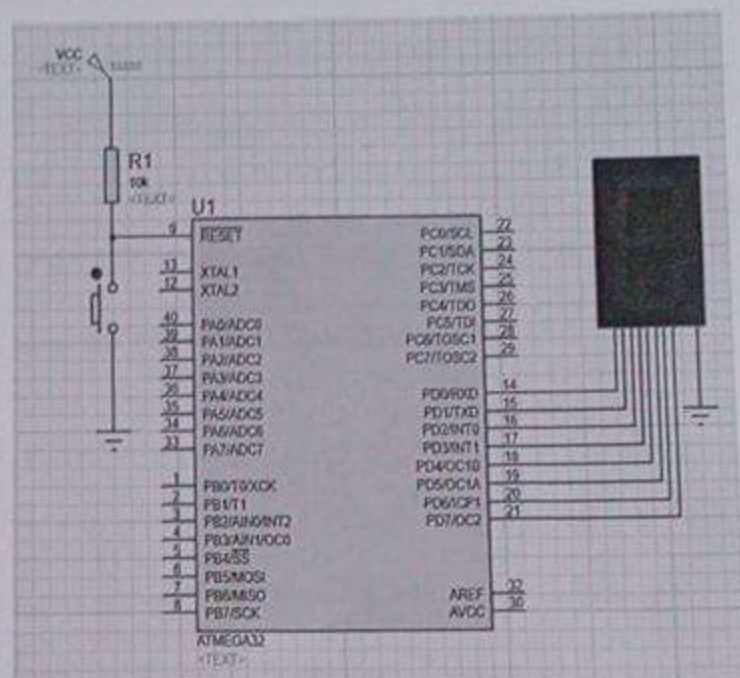


Figure Q5b: 7-Segment display circuit and binary LED table

```

/*7_segment.c
*/
#define F_CPU 8000000UL
#include <avr/io.h>
#include <util/delay.h>
#define LED_Dir DDRD /*define LED Directory*/
#define LED_Port PORTD /*define LED PORT*/

```

```
/*Function declaration*/  
int main(void)  
{  
    /*initialization*/  
    while (1){  
        /*Main loop code here*/  
    }  
}  
/*Function definition*/  
}
```

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